

Dadmehr Daniel Ghasemfar

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TECHNICAL SKILLS

Languages: Java, Python, C++, C, MATLAB, Maple, TensorFlow, HTML/CSS/JavaScript, ROS/ROS2, SQL
Software: CAD (SolidWorks, Fusion, OnShape), FEA and CFD (Ansys, SolidWorks, In-House), PCB (Eagle, Altium), LabVIEW, Excel, Gazebo, Git
Equipment: 3D Printing (SDF, SLS, Metal), Machining (Milling, CNC, Lathe, Woodwork), Lasercutting, Soldering
Theoretical: PDEs, ODEs, Numerical Analysis, GPU Simulations, PID and Controls, Structural and Thermal Analysis, Transfer Functions, Systems Integration
Soft Skills: Project Lead, IP Strategy, Investor Negotiations, Stakeholder Presentations, Customer Engagement, Fluent Farsi, Conversational Spanish

EDUCATION

Duke University Durham, NC
B.S. in Mechanical Engineering, Minors in Math and Psychology Aug 2020 – Dec 2024

- Engineering GPA: 3.6, Math GPA: 3.5
- Courses: Advanced Robotics, Dynamics, Advanced Controls, Heat and Mass Transfer, Fluid Mechanics, Algorithms, Data Structures, ODE/PDEs, Numerical Analysis, Aircraft Design, Chaos Theory, PDE's
- Graduated as a Pratt Fellow for recognition of a two-year research project.

EXPERIENCE

Atar Energy Solutions [thinktank] Aug 2024 – Present
Co-Founder Alexandria, VA

- Designed and prototyped an integrated high-voltage tester, reducing cost and weight by 67% versus competitors, capturing a share in a \$700M market.
- Led business strategy, investor relations, and IP negotiations; finalized patent and corporate formation.
- Co-authored a 120-page technical datasheet and a 70-page professional plan.

Curious Cardinals Mar 2025 – Present
Engineering Mentor Alexandria, VA

- Mentor 10 high school students on 6-month engineering projects (robotic arms, websites, go-karts, etc.), securing 40+ client offers through technical expertise and communication.
- Coordinate with students and parents to align project goals, manage expectations, and ensure successful outcomes.

Duke General Robotics Lab Aug 2022 – May 2024
Robotics Researcher Durham, NC

- Built a GPU-accelerated wave propagation simulator for arbitrary 3D materials and lattices, generating synthetic datasets for AI-based acoustic perception.
- Developed a novel validation method using Chladni plate analysis, bridging simulation and experimental data with 90% accuracy to train an AI model for acoustic perception.
- Research approved by Duke Engineering Board for *Graduation with Distinction award*.

FPrin LLC Oct 2020 – Feb 2021
Lab Technician Mountain View, CA

- Designed, built, and operated case-specific test devices to effectively characterize a recently patented German compressed gas-powered insulin pen. After that, stress tested thousands of components and maintained the shop/lab.
- Engineered custom test fixtures for a patented medical device, enabling the successful characterization and stress-testing of thousands of components to ensure product reliability and performance.

PROJECTS

- Led a 6-person team designing and testing an electro-resistive thruster (162s ISP, 1.36N thrust); submitted results to NASA.
- Designed and fabricated a 2-foot-tall, 8-DOF bipedal walking and dancing robot; developed a Python-based control system for dynamic movement, resolving complex hardware integration issues like mechanical instability.
- Received Mayor's Medal for CNN-based forest fire detection (94% accuracy) using TensorFlow.
- FRC Robotics Quarter-finalist.
- 2x Synopsys Science Fair Finalist (Engineering and CS).
- Received Mayor's Medal for